

ABHIRAM KIDAMBI

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Science in Electrical Engineering and Computer Science

08/24 - Present

- Relevant Courses: Multivariate Calculus, MATLAB Coding, CAD Software, Computer Programs, Data Structures
- GPA: 3.8

Arcadia High School

Arcadia, CA

08/2020 - 06/2024

- GPA: 4.37
- Courses: AP Computer Science A, AP Computer Science Principles, AP Calculus BC, Engineering Design & Development, AP Chemistry

Skills: Java, Python, SQL, Scheme, SolidWorks, MATLAB, Soldering, Tig & Mig Welding, CAD (Computer Aided Design)

INTERNSHIPS

USC SHINE

Los Angeles, CA

RESEARCH INTERN

06/2023 - 08/2023

- Collaborated with Professor Amin Ghadami and a Ph.D. researcher to analyze stability in bridges, vehicles, and aircraft
- Applied engineering principles to study nonlinear aeroelastic flutter instability in structural systems
- Utilized MATLAB to develop models simulating real-world scenarios of structural and aerodynamic instability

ACADEMIC TEAMS

Engineering Design & Development Team for FIRST Robotics

Arcadia, CA

MECHANICAL TEAM MEMBER

08/2022 - 06/2024

- Built a robot to compete in the FIRST Robotics, and JPL Engineering Competitions
- Designed and implemented a pulley system for an extending arm to transport objects to designated locations
- Competed in JPL (Jet Propulsion Laboratory) Invention Challenge and placed top 10 consistently against 40 teams

Arcadia Competition Math Team

Arcadia, CA

Team Member

08/2022 - 09/2023

- Attended team events to work on challenging math problems with other members
- Collaborated with peers, and represented Arcadia High School in regional and national competitions.

Arcadia Applied Engineering (Solar Regatta)

Arcadia, CA

MECHANICAL, DESIGN, AND TECHNICAL TEAM MEMBER

08/2021 - 07/2022

- Part of an award-winning team of ~30 students to build and construct a solar-powered boat
- Oversaw the design of the boat's frame and assisted in material selection/building for the foam hull
- Reinforced the hull with the use of fiberglass and welded the boat's metal frame to ensure structural integrity and durability

PROJECTS

Computer Aided Typing Software:

08/2024 - 12/2024

- Designed a Python application to evaluate typing proficiency using randomly generated typing challenges
- Implemented real-time performance tracking: WPM, accuracy, and elapsed time
- Integrated optional auto-correction and user-personalized text generation based on topic preferences
- Applied modular programming and input handling techniques to ensure responsive interaction

WordNet Graph Processor (Google NGram):

03/2025 - 04/2025

- Parsed WordNet lexical data to construct a directed graph modeling synonym and hyponym relationships
- Built semantic search functionality using graph traversal algorithms and set operations
- Structured the system into parsing, graph construction, and querying modules with high cohesion
- Developed and tested a public API for word relationship queries using JUnit and edge case validation

Tile-Based Game Engine:

04/2025 - 05/2024

- Built a tile-based Java game with procedurally generated worlds using consistent seed-based randomization
- Designed gameplay logic including movement, rendering, and interactive tiles (hoops and locked doors)
- Implemented multi-level transitions and persistent game states using object-oriented principles
- Separated game engine components for extensibility, improving modularity and reusability